

Genetic report for longevity

Circadian Rhythm and Sleep

Molecular clock, chronotype, sleep duration, insomnia, restless legs, melatonin and caffeine

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The circadian clock controls the rhythm of DNA repair, autophagy, glucose and glymphatic brain cleaning; sleeping little or outside hours accelerates several aging mechanisms. This report reads the clock genes (CLOCK, PER1-3, CRY1/2, ARNTL), chronotype, sleep duration, insomnia and sleepiness, restless legs (MEIS1, BTBD9), melatonin MTNR1B receptor and caffeine and melatonin metabolism.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

28

variants

21

in the chip

5

imputed
($DR2 \geq 0.8$)

2

not found

11

Scores with
percentile

Overall reading. The clock is genetically morning (morning chronotype at the 86th percentile and afternoon in the 23), without the rare variant of CRY1 that delays sleep, and with less genetic sensitivity of caffeine to sleep (ADORA2A without the T allele). Being morning favors glucose, mood and sleep regularity. Sleep duration gives opposite signs: the score of 78 loci puts it at the 80th percentile (longest sleep), that of 40 037 variants in 17 (shortest), and PAX8 has 1 short sleep allele; in practice, the need for sleep is typical. The most unfavourable point is insomnia: the 89th percentile in the 129-loci score (63 in the broad score; consensus of the insomnia scores at 80); daytime sleepiness is on the good side (32 and the 1st percentile). In the restless legs, the loci score is at the 77th percentile and MAP2K5/SKOR1 and PTPRD have risk alleles, but the strongest locus (MEIS1) does not have the risk allele. In melatonin, it carries the G allele of MTNR1B: melatonin locks more insulin, and therefore the high dose (especially close to meals) is the most relevant gene-supplement interaction of this report; CYP1A2 *1F heterozygote gives an intermediate clearance of caffeine and melatonin.

Favourable

- **Morning Chronotype (313 loci):** the 86th percentile
- **Chronotype (61 542 variants):** the 23rd percentile
- **Sleep duration (78 loci):** the 80th percentile
- **Diurnal sleepiness (GWAS loci):** the 1st percentile
- **CRY1 rs184039278 TT:** c.1657+3A>C: a rare dominant variant that extends the clock and causes delayed family sleep (sleep and wake up very late).
- **ADORA2A rs5751876 CC:** 1976C>T: T allele is associated with more disturbed sleep and anxiety with caffeine.
- **TOX3 rs78593025 CC:** TOX3: restless legs locus.

Unfavourable

- **Sleep duration (40 037 variants):** the 17th percentile
- **Insomnia (129 loci):** the 89th percentile; consensus p80
- **Restless legs:** the 77th percentile
- **PAX8 rs62158211 GT:** PAX8: one of the most replicated sleep loci.
- **MAP2K5/SKOR1 rs12593813 GG:** Locus MAP2K5/SKOR1 (15q23): sense of risk allele by GWAS.
- **PTPRD rs1975197 GA:** PTPRD: locus of restless legs; felt by GWAS.
- **MTNR1B rs10830963 CG:** Melatonin 2 receptor: G allele increases the expression of the receptor in beta cells; melatonin (natural or supplement) locks more insulin, rises fasting glucose and the risk of type 2 diabetes.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 21 chip genotypes, 0 needed complement, 2 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Molecular clock and chronology

The clock core is a transcription cycle: CLOCK and BMAL1 (ARNTL) bind the genes PER and CRY, whose proteins hold CLOCK/BMAL1 in about 24 h. Variants in these genes and in dozens of loci of GWAS (RGS16, PER2, FBXL13, AK5, VIP) displace the chronotype (mature or afternoon).

Ageing mechanisms: Circadian disorders; Desynchronization of Biological Clocks; Epigenetic Changes

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Morning Chronotype (313 loci)	Favourable	the 86th percentile · z=1.00	Higher score = more morning (more favourable to metabolic health and mood). PGS000336 (PGS Catalog; Genome-wide signifiers; Chronotype; PGP000103, PMID 32579418). Variants with genotype: 305 of 313 (covering of 96.9% coefficients).
Chronotype (61 542 variants)	Favourable	the 23rd percentile · z=-0.77	Higher score = more evening. PGS001992 (PGS Catalog; Penalized regression (bigstatsr); Morning/evening person (chronotype); PGP000263, PMID 34995502). Variants with genotype: 59424 of 61542 (covering of 96.5% coefficients).

Favourable

- Morning Chronotype (313 loci) (the 86th percentile)
- Chronotype (61 542 variants) (the 23rd percentile)
- CRY1 rs184039278 TT

Unfavourable

None.

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CRY1	rs184039278 12:106992962 · T/C/G	TT	chip	G (0; 0%)	strong	c.1657+3A>C: a rare dominant variant that extends the clock and causes delayed family sleep (sleep and wake up very late).

Typical / intermediate variants (5)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CLOCK	rs1801260 4:55435202 · A/G	AG	chip	G (1; 30%)	weak	3111T>C (3'UTR): the C allele (G in strand +) was associated with a more evening chronotype and less sleep in small studies.
RGS16	rs1144566 1:182600491 · T/A/C/G	CC	imputation DR2=0.99	T (0; 2%)	moderate	RGS16: the strongest chronotype locus; cAMP signal in the suprachiasmatic nucleus. Meaning: GWAS Catalog: Morning person (Jones SE 2019; p=2e-95); allele T ↑
FBXL13	rs6465876 7:102857122 · G/A/T	GT	imputation DR2=1.00	T (1; 34%)	moderate	FBXL13: degradation of CRY proteins. Meaning: GWAS Catalog: Morning person (Jones SE 2019; p=4e-24); allele T ↑
AK5	rs12040629 1:77239680 · G/A	GG	chip	A (0; 15%)	moderate	AK5: chronotype locus. Meaning: GWAS Catalog: Morning person (Jones SE 2019; p=5e-45); Allele A ↑
HCRTR2	rs2653349 6:55277539 · A/C/G/T	GG	chip	A (0; 18%)	moderate	HCRTR2: orexin receptor (wake). Meaning: GWAS Catalog: Morning person (Jones SE 2019; p=3e-48); Allele A ↑

Non-judgmental variants (informative) (7)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CLOCK	rs12649507 4:55514317 · G/A/T	AG	chip	—	weak	Intronic CLOCK; associated with sleep duration in some studies.
PER2	rs2304672 2:238277948 · G/C	GG	chip	—	weak	C111G (5'UTR): the G allele was associated with morning preference. Information.
PER3	rs228697 1:7827519 · C/ G	CC	chip	—	weak	P864A: associated with chronotype and seasonal mood.
PER1	rs7221412 17:8137696 · A/G	AG	imputation DR2=0.94	—	weak	PER1 (promoter): associated with waking time and until the day of death.
CRY2	rs10838524 11:45848626 · A/G	GG	chip	—	weak	CRY2: clock repressor; associated with seasonal mood and fasting glucose.
ARNTL	rs2290035 11:13386224 · T/A/C	AA	chip	—	weak	ARNTL (BMAL1): central clock activator; variants associated with sleep and glucose in small studies.
BHLHE41	rs121912617 12:26122364 · G/A/C/T	GG	chip	—	moderate	DEC2 P384R: a rare variant of "short natural sleeper" (less than 1-2 h of sleep without apparent impairment). If absent, the need for sleep is typical.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
VIP	— — · —	—	SNP not found in the data	—	moderate	VIP: neuropeptide that synchronizes the central clock neurons. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Duration and quality of sleep

The duration of sleep is about 10–20% hereditary. Deep sleep (slow waves) is when the glymphatic system cleans the brain more; ADA and the A2A adenosine receptor (ADORA2A) move in the sleep pressure and sensitivity to caffeine.

Ageing mechanisms: Sleep disturbance; Glymphatic dysfunction; Neuroinflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Sleep duration (78 loci)	Favourable	the 80th percentile · $z=0.80$	Higher score : longer sleep. Short sleep is the unfavourable side. PGS003764 (PGS Catalog; Genome-wide signifier SNPs; Sleep duration; PGP000487, PMID 37384397). Variants with genotype: 77 of 78 (covering of 98.6% coefficients).
Sleep duration (40 037 variants)	Unfavourable	the 17th percentile · $z=-0.99$	Higher score : longer sleep. PGS001978 (PGS Catalog; Penalized regression (bigstatsr); Sleep duration; PGP000263, PMID 34995502). Variants with genotype: 38799 of 40037 (covering of 97.0% coefficients).
Short sleep (≤ 6 h)	SNP not found in the data	—	Higher score = more tendency for short sleep.

Favourable

- Sleep duration (78 loci) (the 80th percentile)
- ADORA2A rs5751876 CC

Unfavourable

- Sleep duration (40 037 variants) (the 17th percentile)
- PAX8 rs62158211 GT

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ADORA2A	rs5751876 22:24441333 · T/C	CC	chip	T (0; 39%)	moderate	1976C>T: T allele is associated with more disturbed sleep and anxiety with caffeine.

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
PAX8	rs62158211 2:113348562 · G/T	GT	imputation DR2=0.99	T (1; 21%)	moderate	PAX8: one of the most replicated sleep loci. Meaning: GWAS Catalog: Sleep duration (Jones SE 2016; $p=2e-23$); G ↓ allele

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ADA	rs73598374 20:44651586 · C/A/G/T	CC	chip	C (2; 94%)	moderate	G22N: the rare allele 22N (T in strand +) lowers adenosine deaminase activity and increases deep sleep and sleep pressure; the common allele is C.

Insomnia and daytime sleepiness

Insomnia shares with anxiety and depression (MEIS1 is the strongest locus). Daytime sleepiness and sleep apnea are close themes.

Ageing mechanisms: Sleep disturbance; Endocrine dysregulation; Neuroinflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Insomnia (129 loci)	Unfavourable	the 89th percentile · z=1.16 · consensus p80	Higher score : higher risk of insomnia. PGS003859 (PGS Catalog; Genome-wide signifier SNPs; Insomnia; PGP000497, PMID 37013595). Variants with genotype: 123 of 129 (covering of 90.4% coefficients).
Insomnia (37 712 variants)	Typical / intermediate	the 63rd percentile · z=0.32	Higher score : higher risk. PGS001932 (PGS Catalog; Penalized regression (bigstatsr); Sleepiness / insomnia; PGP000263, PMID 34995502). Variants with genotype: 36455 of 37712 (covering of 96.6% coefficients).
Diurnal sleepiness (27 143 variants)	Typical / intermediate	the 32nd percentile · z=-0.51	Higher score = more daytime sleepiness. PGS001995 (PGS Catalog; Penalized regression (bigstatsr); Daytime dozing / sleeping (narcolepsy); PGP000263, PMID 34995502). Variants with genotype: 26383 of 27143 (covering of 97.3% coefficients).
Obstructive sleep apnea (18 loci)	Typical / intermediate	the 43rd percentile · z=-0.18	Higher score : higher risk. PGS003857 (PGS Catalog; Genome-wide signifier SNPs; Obstructive sleep apnea; PGP000496, PMID 36989840). Variants with genotype: 17 of 18 (covering of 93.8% coefficients).
Diurnal sets	Typical / intermediate	the 36th percentile · z=-0.36	Higher score = more naps. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 91 of 96 loci with genotype. Studies: Dashti HS (Daytime nap); Hammerschlag AR (Daytime nap); Jansen PR (Daytime nap). Variants with genotype: 91 of 96 (covering of 94.8% coefficients).
Diurnal sleepiness (GWAS loci)	Favourable	the 1st percentile · z=-2.66	Higher score = more sleepiness. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 8 of 11 loci with genotype. Studies: Hammerschlag AR (Daytime sleepiness); Jansen PR (Daytime sleepiness); Lane JM (Excessive day sleepiness); Nagarajan P (Apnea hypopnea index x excellent day sleepiness interaction (1df)). Variants with genotype: 8 of 11 (covering of 72.7% coefficients).

Favourable

- Diurnal sleepiness (GWAS loci) (the 1st percentile)

Unfavourable

- Insomnia (129 loci) (the 89th percentile; consensus p80)

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MED27	rs7040224 · 9:132011450 · A/G	GA	chip	A (1; 31%)	moderate	MED27: insomnia locus. Meaning: GWAS Catalog: Insomnia (Jansen PR 2019; p=4e-09); Allele A ↑

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CACNA1A	— · —	—	SNP not found in the data	—	moderate	CACNA1A: neuronal calcium channel; insomnia locus. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Restless legs syndrome

The restless legs worsen the fall asleep and the deep sleep. Strong loci: MEIS1, BTBD9, MAP2K5/SKOR1, PTPRD, TOX3. Low brain iron is the main treatable factor.

Ageing mechanisms: Sleep disturbance; Neuromuscular Junction Disorder

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Restless legs	Unfavourable	the 77th percentile · z=0.79	Higher score : higher risk. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 175 of 195 loci with genotype. Studies: Lin GY (Restless leg syndrome in migraine without aura); Schormair B (Restless leg syndrome). Variants with genotype: 175 of 195 (covering of 89.7% coefficients).

Favourable

- TOX3 rs78593025 CC

Unfavourable

- Restless legs (the 77th percentile)
- MAP2K5/SKOR1 rs12593813 GG
- PTPRD rs1975197 GA

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TOX3	rs78593025 · 16:52485054 · C/A/T	CC	imputation · DR2=0.99	T (0; 44%)	moderate	TOX3: restless legs locus. Meaning: GWAS Catalog: Restless leg syndrome (Schormair B 2024; p=2e-66); T ↑ allele

Unfavourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MAP2K5/SKOR1	rs12593813 · 15:67744514 · A/G	GG	chip	G (2; 64%)	strong	Locus MAP2K5/SKOR1 (15q23): sense of risk allele by GWAS. Meaning: GWAS Catalog: Restless leg syndrome (Winkelmann J 2011; p=1e-22); G ↑ allele
PTPRD	rs1975197 · 9:8846955 · G/A	GA	chip	A (1; 17%)	moderate	PTPRD: locus of restless legs; felt by GWAS. Meaning: GWAS Catalog: Restless leg syndrome (Winkelmann J 2011; p=3e-10); Allele A ↑

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MEIS1	rs2300478 · 2:66554321 · T/G	TT	chip	G (0; 25%)	strong	MEIS1 (8 intron): G allele is the main common factor of restless legs (OR ~1.7) and also increases insomnia.
BTBD9	rs3923809 · 6:38473194 · A/G	GA	chip	A (1; 69%)	strong	BTBD9: Allele A increases the restless legs and periodic movements of the limbs; associated with lower ferritin.

Melatonin, glucose and caffeine

MTNR1B is the melatonin receptor in pancreatic beta cells: G allele increases the receptor and melatonin stops more insulin. CYP1A2 metabolizes caffeine and melatonin itself.

Ageing mechanisms: Circadian disorders; Hyperglycemia and Intolerance to Glucose; Insulin resistance

Favourable

None.

Unfavourable

- MTNR1B rs10830963 CG

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MTNR1B	rs10830963 11:92975544 · C/G/T	CG	chip	G (1; 29%)	strong	Melatonin 2 receptor: G allele increases the expression of the receptor in beta cells; melatonin (natural or supplement) locks more insulin, rises fasting glucose and the risk of type 2 diabetes. Classified as unfavourable because it is carrying the G allele (1 copy): it is the genotype of interaction with melatonin at a high dose.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CYP1A2	rs762551 15:74749576 · C/A/G	AC	chip	C (1; 32%)	moderate	CYP1A2*1F (-163C>A): allele A makes the enzyme more inducible (fast purification of caffeine and melatonin); C = slower depuration.

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
AHR	rs4410790 7:17244953 · T/C	CC	chip	—	moderate	AHR: regulates CYP1A2; the allele that increases coffee consumption accelerates caffeine clearance.
CYP2C19	rs12248560 10:94761900 · C/A/T	CC	chip	—	moderate	CYP2C19*17: secondary route of melatonin; informative (see pharmacogenomic).

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Sleep disturbance	Sleep duration (40 037 variants) (the 17th percentile); PAX8 GT; Insomnia (129 loci) (the 89th percentile; consensus p80); Restless legs (the 77th percentile); MAP2K5/SKOR1 GG; PTPRD GA	Sleep duration (78 loci) (the 80th percentile); ADORA2A CC; Diurnal sleepiness (GWAS loci) (the 1st percentile); TOX3 CC	ADA CC; Insomnia (37 712 variants) (the 63rd percentile); Diurnal sleepiness (27 143 variants) (the 32nd percentile); Obstructive sleep apnea (18 loci) (the 43rd percentile); Diurnal sections (the 36th percentile); MED27 GA; MEIS1 TT; BTBD9 GA
Neuromuscular Junction Disorder	Restless legs (the 77th percentile); MAP2K5/SKOR1 GG; PTPRD GA	TOX3 CC	MEIS1 TT; BTBD9 GA
Neuroinflammation	Sleep duration (40 037 variants) (the 17th percentile); PAX8 GT; Insomnia (129 loci) (the 89th percentile; consensus p80)	Sleep duration (78 loci) (the 80th percentile); ADORA2A CC; Diurnal sleepiness (GWAS loci) (the 1st percentile)	ADA CC; Insomnia (37 712 variants) (the 63rd percentile); Diurnal sleepiness (27 143 variants) (the 32nd percentile); Obstructive sleep apnea (18 loci) (the 43rd percentile); Diurnal sections (the 36th percentile); MED27 GA
Glymphatic dysfunction	Sleep duration (40 037 variants) (the 17th percentile); PAX8 GT	Sleep duration (78 loci) (the 80th percentile); ADORA2A CC	ADA CC
Endocrine dysregulation	Insomnia (129 loci) (the 89th percentile; consensus p80)	Diurnal sleepiness (GWAS loci) (the 1st percentile)	Insomnia (37 712 variants) (the 63rd percentile); Diurnal sleepiness (27 143 variants) (the 32nd percentile); Obstructive sleep apnea (18 loci) (the 43rd percentile); Diurnal sections (the 36th percentile); MED27 GA
Hyperglycemia and Intolerance to Glucose	MTNR1B CG	—	CYP1A2 AC
Circadian disorders	MTNR1B CG	Morning Chronotype (313 loci) (the 86th percentile); Chronotype (61 542 variants) (the 23rd percentile); CRY1 TT	RGS16 CC; FBXL13 GT; AK5 GG; HCRTR2 GG; CYP1A2 AC
Insulin resistance	MTNR1B CG	—	CYP1A2 AC
Epigenetic Changes	—	Morning Chronotype (313 loci) (the 86th percentile); Chronotype (61 542 variants) (the 23rd percentile); CRY1 TT	RGS16 CC; FBXL13 GT; AK5 GG; HCRTR2 GG
Desynchronization of Biological Clocks	—	Morning Chronotype (313 loci) (the 86th percentile); Chronotype (61 542 variants) (the 23rd percentile); CRY1 TT	RGS16 CC; FBXL13 GT; AK5 GG; HCRTR2 GG

Practical part

Ideas for sleep and clock, crossed with the genetic plan v2 and current medication (isotretinoin, rapamycin, lithium 5 mg, tadalafil, dapagliflozin, retatrutide, telmisartan, LDN 4.5 mg, melatonin). The current stack remains as it is; everything is suggestion (not applied). Genetics points to three priorities: protecting sleep from insomnia (high score), correcting the dose and time of melatonin (MTNR1B G) and maintaining the regularity of a clock that is already morning.

1. Melatonin: down to 0.3–1 mg and away from meals

Genetics: MTNR1B rs10830963 C/G; CYP1A2 *1F A/C

Physiological doses (0.3–1 mg) 30–60 min before bedtime hit the clock as well as high doses, without leaving melatonin high in the morning. Taking ≥ 3 h after the last meal: in carrier G, melatonin next to food rises glucose. If you want to keep a high dose for other reasons, never close to meals and without breakfast very early.

V2 genetic plan and medication: Similar to the v2 plan (S30: up to 100 mg for 0.3–1 mg). • suggestion (not applied)

2. Caffeine: ≤ 200 mg and only until 12h

Genetics: Insomnia p80; CYP1A2 *1F A/C (intermediate purification); ADORA2A CC (less sensitive)

Even with favourable ADORA2A, the half-life of caffeine (5–6 h, a little more with intermediate CYP1A2) arrives at night. Filtered coffee, only in the morning.

V2 genetic plan and medication: Like the v2 plan (item 10). • suggestion (not applied)

3. Light: strong morning, dark night

Genetics: Morning Chronotype (p86); clock without phase delay

10–30 min of outdoor light in the first hour after waking; hot and low light in the 2 h before bedtime; dark room. Time to wake up fixed (± 30 min), also at the weekend: it is the one that stabilizes the clock the most.

V2 genetic plan and medication: Compatible with the plan. • suggestion (not applied)

4. Food schedule

Genetics: MTNR1B G; morning chronotype

Food window of 10–12 h, hydrates mainly in the morning and at lunch, last meal ≥ 3 h before bedtime, without night snack.

V2 genetic plan and medication: Similar to the v2 plan (hour). • suggestion (not applied)

5. If insomnia arises: CBT-I first

Genetics: Insomnia p80 on the 129-loci score (consensus p80); MEIS1 without the risk allele

Cognitive-behavioral therapy for insomnia (CBT-I, including validated apps) is the first line treatment and lasts longer than hypnotics. Glycine 3 g at bedtime and magnesium glycinate, already in the plan, help a little. Avoid chronic benzodiazepines and "drugs Z".

V2 genetic plan and medication: Glycine (S8) and magnesium (S13) already in the plan. • suggestion (not applied)

6. Medication and sleep

Genetics: — (drug effect)

Dapagliflozin in the morning (diuresis at night interrupts sleep). If LDN causes vivid dreams or insomnia, it takes to the morning (effectiveness remains in many reports). Lithium slightly stretches the clock period; with 5 mg the effect is minimal.

V2 genetic plan and medication: Compatible with the plan (does not change doses). • suggestion (not applied)

7. Restless legs: watching and knowing what to do

Genetics: Restless legs score p77; MAP2K5/SKOR1 GG; PTPRD GA; BTBD9 GA

If there is an irresistible desire to move the legs at the end of the day, which alleviates when walking: measuring ferritin and transferrin saturation. The restless legs guides ask for ferritin ≥ 75 ; the fixed plan < 100 , so the common target would be 75–100, only if there are symptoms and with the doctor. Avoid sedative antihistamines and late caffeine, which aggravate.

V2 genetic plan and medication: Coherent with the v2 plan (ferritin ≥ 75), with the 75–100 window only if there are symptoms. • suggestion (not applied)

8. Temperature and exercise

Genetics: Sleep duration with opposite signs; deep sleep (ADA without the 22N allele)

Fourth to 17–19 °C; hot shower or bath 1–2 h before bedtime (the temperature decrease below deepens sleep); intense exercise in the morning or afternoon, not in the last 1–2 h before bedtime.

V2 genetic plan and medication: Compatible with the plan. • suggestion (not applied)

9. Measuring sleep, not genetics

Genetics: Contradictory sleep duration scores

A record of 2 weeks (daily or wearable: bedtime and waking time, awakenings, sleepiness) says more about the real need for sleep than scores. If there is strong snoring or pauses in the reported breathing, sleep study (apnea: typical score, p43).

V2 genetic plan and medication: New; no cost. • suggestion (not applied)

Avoiding or with caution

- Melatonin in high dose near meals or at dawn (MTNR1B G). suggestion (not applied)
- Coffee after 12h and energy drinks. suggestion (not applied)
- Screens and strong white light in the 2 h before bedtime. suggestion (not applied)
- Antihistamines sedatives as "compressed to sleep" (aggravated the restless legs and cognition the following day). suggestion (not applied)
- Heavy meals and intense exercise in 2–3 h before bedtime. suggestion (not applied)

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. Structural variants (VNTR of PER3) are not read.
- The clock genes (CLOCK, PER, CRY, ARNTL) have mainly associations of small studies; the chronotype scores, sleep duration and insomnia come from large GWAS, but explain little of the individual variation.
- Sleep depends much more on light, schedules, caffeine, exercise, medication and stress than these genes.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS000336 (PGS Catalog; Genome-wide signifiers; Chronotype; PGP000103, PMID 32579418 · PGS001992 (PGS Catalog; Penalized regression (bigstatsr); Morning/evening person (chronotype); PGP000263, PMID 34995502 · PGS003764 (PGS Catalog; Genome-wide signifier SNPs; Sleep duration; PGP000487, PMID 37384397 · PGS001978 (PGS Catalog; Penalized regression (bigstatsr); Sleep duration; PGP000263, PMID 34995502 · PGS003859 (PGS Catalog; Genome-wide signifier SNPs; Insomnia; PGP000497, PMID 37013595 · PGS001932 (PGS Catalog; Penalized regression (bigstatsr); Sleepiness / insomnia; PGP000263, PMID 34995502 · PGS001995 (PGS Catalog; Penalized regression (bigstatsr); Daytime dozing / sleeping (narcolepsy); PGP000263, PMID 34995502 · PGS003857 (PGS Catalog; Genome-wide signifier SNPs; Obstructive sleep apnea; PGP000496, PMID 36989840).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

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